Cloud-Based Campus Resource Sharing & Utilization Platform

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Synopsis / Abstract

This cloud computing project proposes a 'Cloud-Based Campus Resource Sharing and Utilization Platform' aimed at improving how educational institutions manage and share physical and digital campus resources such as classrooms, laboratories, equipment, and study materials. The platform leverages cloud infrastructure to provide real-time availability tracking, automated booking, resource utilization analytics, and role-based access for students, faculty, and administrators. By centralizing resource management on a scalable cloud platform, the system reduces resource conflicts, minimizes idle capacity, and improves overall efficiency of campus operations.

Problem Statement

Educational campuses often struggle with inefficient utilization and allocation of shared resources such as seminar halls, laboratories, projectors, and sports facilities. Resource booking is frequently handled manually through registers, emails, or disconnected systems, leading to double bookings, underutilized assets, and lack of visibility into real-time availability. Administrators also lack consolidated data on how resources are used over time, making it difficult to plan capacity or identify wastage.

This problem affects students, faculty, and administrative staff across departments who depend on shared campus infrastructure. Addressing it is important to reduce scheduling conflicts, improve transparency, optimize resource usage, and support data-driven decisions for campus resource planning.

Proposed Solution

The proposed solution is a cloud-hosted 'Campus Resource Sharing and Utilization Platform' that centralizes the discovery, booking, and monitoring of campus resources. It consists of five main modules:

User Authentication and Role Management Module: Provides secure sign-up and login for students, faculty, and administrators with role-based access control, enabling appropriate permissions for booking and approval workflows.

Resource Catalog and Booking Module: Maintains a real-time catalog of available resources (classrooms, labs, equipment, and facilities) and allows users to search, check availability, and book resources for specific time slots, with automated conflict detection.

Approval and Notification Module: Routes booking requests through configurable approval workflows and sends automated email or in-app notifications for confirmations, reminders, and cancellations.

Utilization Analytics Module: Tracks resource usage patterns and presents visual dashboards showing utilization rates, peak demand periods, and underused assets to support administrative decision-making.

Cloud Infrastructure and Scalability Module: Hosts the platform on a cloud service provider, using managed services for compute, storage, and database to ensure high availability, scalability, and secure data handling across the institution.

This system addresses the problems of manual, conflict-prone resource booking and lack of utilization visibility by combining real-time scheduling, automated workflows, and cloud-based analytics into a single accessible platform.

Tools and Technologies

Cloud Service Provider: AWS / Microsoft Azure / Google Cloud Platform

Cloud Managed Services: Compute (EC2 / App Service), Object Storage (S3 / Blob Storage), Managed Database Service, Load Balancer, Cloud Notification Service

Databases: MySQL / PostgreSQL (relational data), MongoDB (optional, for logs and analytics)

Source Version Control: Git and GitHub

Other Tools: Docker for containerization, CI/CD pipeline (GitHub Actions), React/Angular for frontend, Node.js/Spring Boot for backend APIs

Expected Outcomes

The expected outcome is a fully functional, cloud-hosted web application that streamlines campus resource booking and utilization tracking. The system will be considered successful if it eliminates double bookings, provides real-time resource availability, and delivers accurate utilization analytics to administrators. Additional success measures include improved user satisfaction through faster booking workflows, reduced manual coordination effort, and demonstrable scalability of the cloud deployment under increased user load.

Tentative Timeline

Week #	Phase/Tasks
1-4	Requirement Analysis & Specification

4-5	High Level Design
5-6	Low Level Design (UI/UX, Backend API, Database Integration, Cloud Architecture)
4-7	Implementation, Unit Testing & Debugging
7-9	Testing (System & Integration Testing & Debugging)
9-10	Deployment & Release on Cloud
11	Documentation (Design Documentation, Test Results, Presentation PPT) Preparation
12	Final Presentation